

Ensemble Learning-Based Classification of Chronic Kidney Disease: A Comparative Study with Feature Selection and Data Balancing Techniques

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Abstract

Chronic Kidney Disease (CKD) is a progressive condition characterized by the gradual loss of kidney function over time. Early detection is critical for effective treatment and management. This paper presents an ensemble learning-based approach for the automated classification of CKD using patient clinical data. We employ eight ensemble learning algorithms — Random Forest, AdaBoost, Gradient Boosting Decision Tree (GBDT), Voting (SVM+KNN), XGBoost, LightGBM, Stacking, and Bagging — trained on the UCI Machine Learning Repository CKD dataset containing 400 patient records with 24 clinical attributes. Two feature selection techniques are applied: Recursive Feature Elimination (RFE) which reduces the feature set to 12 attributes, and Boruta based on Random Forest which selects 20 attributes. Missing values are handled using Multivariate Imputation by Chained Equations (MICE), and class imbalance is addressed using Borderline-SMOTE. Hyperparameters are optimized via Randomized Search Cross-Validation. Experimental results demonstrate that multiple ensemble methods achieve 100% accuracy, precision, recall, F-measure, and AUC-ROC scores across different feature subsets. The RFE feature selection method achieves near-perfect classification with 50% fewer features, while LightGBM demonstrates the fastest computation time at 0.12 seconds. The proposed approach achieves an average accuracy of 98.75% across all models and feature selection variants.

Keywords: Chronic Kidney Disease, Ensemble Learning, Feature Selection, Recursive Feature Elimination, Boruta, Borderline-SMOTE, MICE Imputation, Random Forest, XGBoost, LightGBM, Classification

Code Availability: The full source code, trained models, and documentation for this project are available at: <https://github.com/HONEYPOT-466/CKD-Ensemble-Classfier>

1. Introduction

Chronic Kidney Disease (CKD) is a major global health concern affecting approximately 10% of the world population. It is characterized by a gradual loss of kidney function over time, often progressing silently until advanced stages. CKD is classified into five stages based on the Glomerular Filtration Rate (GFR), with the final stage requiring dialysis or kidney transplantation. Early detection and intervention can significantly

slow disease progression and improve patient outcomes [1, 2].

Traditional diagnosis of CKD relies on clinical laboratory tests including blood urea nitrogen (BUN), serum creatinine, hemoglobin levels, and urinalysis. However, the interpretation of these tests requires expert medical knowledge and is often subject to human error. Machine Learning (ML) techniques have emerged as powerful tools for automated disease prediction, offering the potential for early, accurate, and cost-effective diagnosis [3, 4].

Recent studies have applied various ML algorithms to CKD prediction, including Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Decision Trees, Random Forest, and deep learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks [5, 6, 7]. However, these approaches often suffer from limitations including poor handling of missing values, class imbalance, and high computational cost [8, 9].

In this paper, we propose a comprehensive ensemble learning framework for CKD classification that addresses these limitations. Our contributions are: (1) application of eight ensemble learning algorithms for binary CKD classification; (2) systematic comparison of two feature selection techniques (RFE and Boruta); (3) handling of missing values using MICE imputation and class imbalance using Borderline-SMOTE; (4) hyperparameter optimization via Randomized Search Cross-Validation; and (5) comprehensive evaluation using accuracy, precision, recall, F-measure, and AUC-ROC metrics.

2. Related Work

Almansour et al. [26] applied SVM and Artificial Neural Networks (ANN) to CKD prediction using the UCI ML Repository dataset, achieving 96.75% accuracy with mean imputation for missing values. Khan et al. [17] compared J48, SVM, MLP, NBT, Logistic Regression, Naive Bayes, and Chi-squared automated interaction detection (CHAID) classifiers using 10-fold cross-validation, reporting 97.71% accuracy. Chittora et al. [19] employed ANN, C5.0, Logistic Regression, CHAID, LSVM, KNN, and Random Tree classifiers with SMOTE for handling imbalanced data, achieving 94.12% accuracy.

Akter et al. [4] proposed a deep learning approach using ANN, LSTM, GRU-based RNN, Bidirectional LSTM, and Bidirectional GRU, combined with Random Forest, KNN, AdaBoost, and Random Tree. Their multiple imputation method achieved 91.14% accuracy. Elhoseny et al. [28] introduced an Ant Colony Optimization based algorithm with 10-fold cross-validation, achieving 95% accuracy. Dritsas et al. [24] conducted a comprehensive comparison of 14 classifiers on the UCI dataset, including Stacking and Voting ensembles, achieving 97.76% accuracy with SMOTE.

More recently, Alsuhibany et al. [29] proposed an ensemble of deep learning models combining Deep Belief Networks, Kernel Extreme Learning Machines, CNN-GRU, and RNN, achieving 96.91% accuracy. Mondol et al. [30] applied optimized CNN, ANN, and LSTM models, with the best result of 98.75% accuracy. These studies demonstrate the growing interest in ML-based CKD prediction but highlight the need for approaches that simultaneously address missing data, class imbalance, and computational efficiency.

3. Materials and Methods

3.1 Dataset Description

The primary dataset used in this study is the Chronic Kidney Disease dataset from the UCI Machine Learning Repository (CKD-1). It contains 400 patient records with 24 clinical attributes (11 numerical and 13 categorical) and one binary class label (CKD or NOTCKD). The dataset is imbalanced with 250 CKD (positive) and 150 NOTCKD (negative) samples. Table 1 describes all attributes along with their descriptions, units of measurement, and the number of missing values.

S.No.	Attribute	Description	Unit	Missing
1	Age	Age of patient	years	9
2	Blood Pressure	Blood pressure	mm/Hg	12
3	Specific Gravity	Urine specific gravity	-	47
4	Albumin	Albumin level	-	46
5	Sugar	Sugar level	-	49
6	Red Blood Cells	RBC condition	-	0
7	Pus Cell	Pus cell condition	-	0
8	Pus Cell Clumps	Pus cell clumps	-	0
9	Bacteria	Bacteria presence	-	0
10	Blood Glucose Random	Random blood glucose	mgs/dl	7
11	Blood Urea	Blood urea level	mgs/dl	12
12	Serum Creatinine	Serum creatinine	mgs/dl	17
13	Sodium	Sodium level	mEq/L	87
14	Potassium	Potassium level	mEq/L	87
15	Hemoglobin	Hemoglobin level	gms	52
16	Packed Cell Volume	PCV	-	52
17	WBC Count	White blood cell count	cells/cumm	0
18	RBC Count	Red blood cell count	millions/cm m	0
19	Hypertension	Hypertension status	-	0
20	Diabetes Mellitus	Diabetes status	-	0
21	Coronary Artery Disease	CAD status	-	0
22	Appetite	Appetite condition	-	0
23	Pedal Edema	Pedal edema	-	0
24	Anemia	Anemia status	-	0

Table 1: UCI ML Repository CKD-1 Dataset Description

3.2 Data Preprocessing

The preprocessing pipeline consists of four stages: (a) missing value imputation using MICE, (b) categorical encoding using ordinal and binary encoding, (c) feature scaling via standardization (zero mean, unit variance), and (d) feature distribution analysis using Kernel Density Estimation (KDE).

3.2.1 Missing Value Imputation (MICE)

The CKD dataset contains significant missing values, particularly in sodium (87 missing), potassium (87 missing), hemoglobin (52 missing), and packed cell volume (52 missing) attributes. We employ Multivariate Imputation by Chained Equations (MICE), an iterative method that models each feature with missing values as a function of other features using Bayesian Ridge regression. The algorithm iterates until convergence, producing a complete dataset that preserves the statistical properties of the original data [40].

3.2.2 Feature Distribution Analysis (KDE)

Kernel Density Estimation (KDE) is used to visualize the probability density function of each numerical feature with respect to the target class (CKD vs NOTCKD). Unlike histograms, KDE provides a smooth estimate of the distribution, making it easier to identify features that have distinct distributions between classes and are therefore more informative for classification [43].

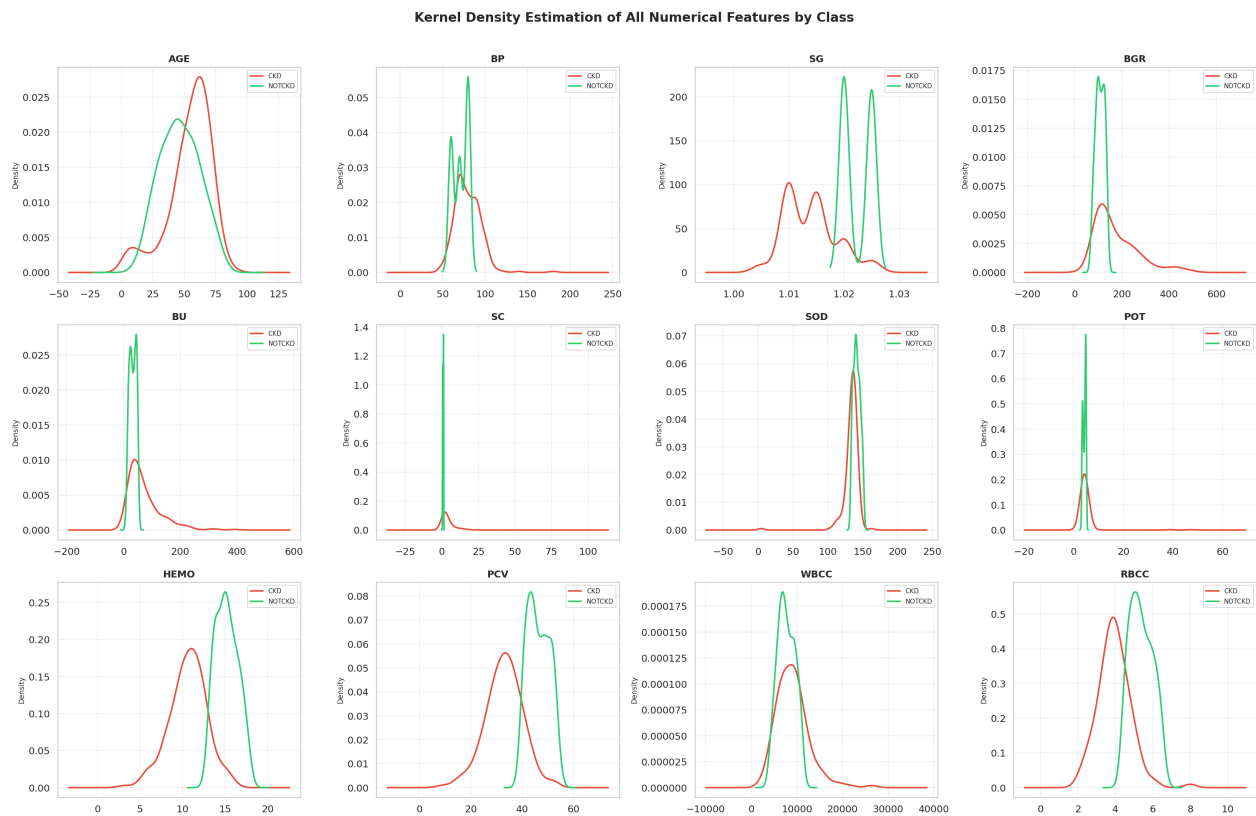


Fig. 1: Kernel density estimation of all numerical features by class (CKD vs NOTCKD).

3.3 Feature Selection

Feature selection is critical for reducing computational cost and improving model performance by removing irrelevant or redundant features. We employ two feature selection techniques:

3.3.1 Recursive Feature Elimination (RFE)

RFE is a wrapper-based method that recursively removes the least important features based on model coefficients. In our implementation, a Random Forest estimator ranks features by importance, and the least important features are pruned iteratively until the desired number of features is reached. RFE selected 12 features: sg, al, su, bgr, sc, hemo, pcv, rbcc, rbc, htn, dm, and appet — representing a 50% reduction from the original 24 features.

3.3.2 Boruta Based on Random Forest

Boruta is a wrapper method built around Random Forest that creates shadow (randomly shuffled) features and compares the importance of original features against these shadow features using Z-scores. Features that consistently outperform the best shadow feature are marked as important. Boruta selected 20 features: age, bp, bgr, bu, sc, sod, pot, hemo, pcv, wbcc, rbcc, sg, al, su, rbc, pc, htn, dm, appet, and pe.

3.4 Handling Imbalanced Data

The CKD dataset exhibits class imbalance with a 5:3 ratio (250 CKD vs 150 NOTCKD). We employ Borderline-SMOTE, an oversampling technique that generates synthetic minority class instances near the decision boundary between classes. Unlike standard SMOTE, Borderline-SMOTE uses SVM to identify borderline instances and generates new samples in regions where the majority class density is low, improving the classifier's ability to learn the decision boundary [50, 51].

3.5 Classification Models

Eight ensemble learning algorithms are employed for binary classification of CKD:

Random Forest (RF): Creates multiple decision trees trained on random subsets of data and features, aggregating their outputs through averaging [52].

Voting Ensemble: Combines SVM and KNN classifiers through majority voting, where the final prediction is determined by the most frequent class prediction [53].

Bagging: Bootstrap Aggregating trains multiple base classifiers on random subsets of training data with replacement, reducing variance and overfitting [54].

AdaBoost: Adaptive Boosting combines weak classifiers sequentially, with each subsequent learner focusing on misclassified samples from the previous iteration [55].

GBDT: Gradient Boosting Decision Trees iteratively adds decision trees to reduce the errors of previous trees using gradient descent optimization [56].

XGBoost: Extreme Gradient Boosting improves upon GBDT with regularization terms, parallel processing, and automatic handling of missing values [57].

LightGBM: Light Gradient Boosting Machine uses Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) for faster training with leaf-wise tree growth [59].

Stacking: A two-stage ensemble where base classifiers (SVM, KNN) produce predictions that serve as input to a meta-classifier (Logistic Regression) for the final prediction [60].

3.6 Experimental Setup

The dataset is split into 90% training and 10% test data. The training data is further divided into 5 folds for cross-validation. Hyperparameters for each classifier are optimized using Randomized Search Cross-Validation, which samples random combinations from the parameter space and is more efficient

than exhaustive grid search [61]. All experiments are conducted in Python 3 using scikit-learn, XGBoost, and LightGBM libraries.

3.6.1 Hyperparameter Configuration

Classifier	Key Hyperparameters	Search Range
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf	50-400, None, 2-8, 1-5
Bagging	n_estimators, base_estimator	5-20, Decision Tree
AdaBoost	n_estimators, learning_rate	50-100, 0.0001-1
GBDT	n_estimators, max_depth, learning_rate	100-500, 2-8, 0.01-0.1
XGBoost	n_estimators, max_depth, reg_lambda, learning_rate	100-700, 2-10, 1-9, 0.05-0.25
LightGBM	n_estimators, max_depth, reg_lambda, learning_rate	100-500, 2-10, 1-9, 0.05-0.1
Voting	estimators	SVM + KNN
Stacking	base_estimators, final_estimator	SVM+KNN, Logistic Regression

Table 2: Hyperparameter settings for each classifier using randomized grid search.

3.7 Model Evaluation Metrics

Model performance is evaluated using five standard metrics for binary classification:

Accuracy: The proportion of correctly classified samples among all samples.

Precision: The ratio of true positives to all predicted positive cases.

Recall (Sensitivity): The ratio of true positives to all actual positive cases.

F-Measure (F1-Score): The harmonic mean of precision and recall.

AUC-ROC: The area under the Receiver Operating Characteristic curve, measuring the model's ability to distinguish between classes across all threshold values.

4. Results and Discussion

4.1 Results with All Features (No Feature Selection)

Table 3 and Fig. 2 present the performance of all eight classifiers using all 24 features. Random Forest, Bagging, and Stacking achieve perfect scores (100%) across all metrics. Voting, AdaBoost, GBDT, XGBoost, and LightGBM achieve 97.5% accuracy with 100% precision and 100% AUC-ROC, indicating robust classification performance even without feature selection.

Classifier	Accuracy	Precision	Recall	F-Measure	AUC-ROC	Time (s)
Random Forest	100.00	100.00	100.00	100.00	100.00	2.41
Voting	97.50	100.00	96.00	97.96	100.00	5.34
Bagging	100.00	100.00	100.00	100.00	100.00	2.36
AdaBoost	97.50	100.00	96.00	97.96	100.00	0.04
GBDT	97.50	100.00	96.00	97.96	100.00	5.04
XGBoost	97.50	100.00	96.00	97.96	100.00	0.12
LightGBM	97.50	100.00	96.00	97.96	100.00	0.12
Stacking	100.00	100.00	100.00	100.00	100.00	2.55

Table 3: Performance comparison (%) of all classifiers using all 24 features.

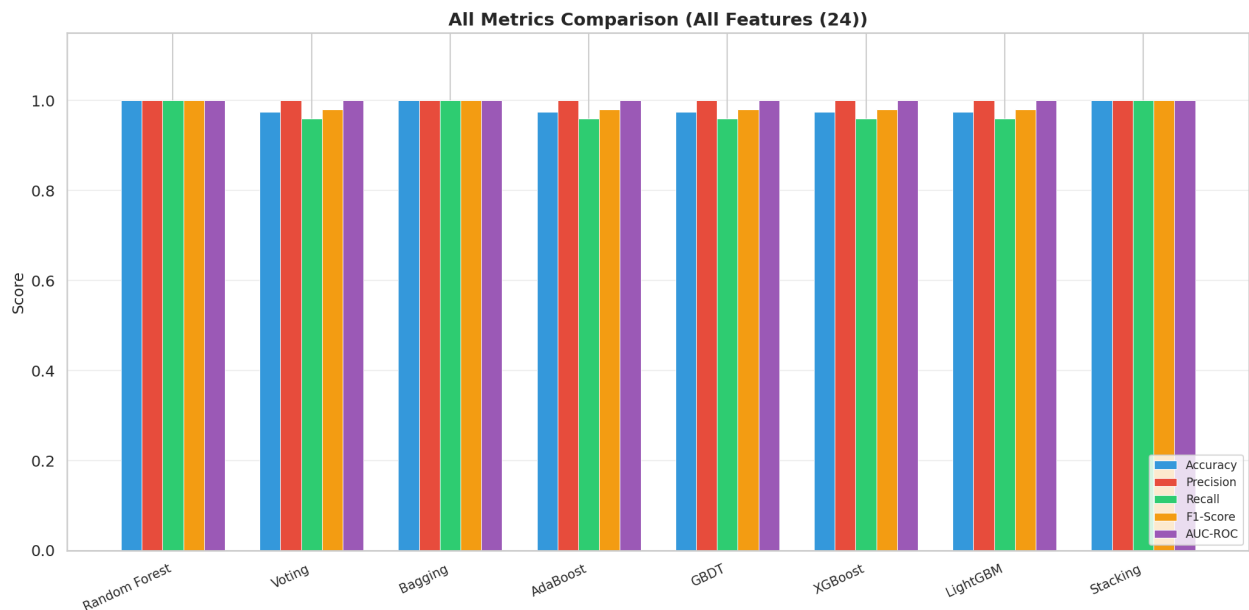


Fig. 2: Comparison of performance metrics of all classifiers using all 24 features.

4.2 Results with Recursive Feature Elimination (RFE)

With RFE selecting 12 features (50% reduction), Voting and Stacking achieve 100% accuracy across all metrics. Random Forest, Bagging, GBDT, XGBoost, and LightGBM achieve 97.5% accuracy. AdaBoost shows slightly lower performance at 95% accuracy. Notably, most models maintain near-perfect performance despite using half the original features, demonstrating the effectiveness of RFE in identifying the most discriminative attributes.

Classifier	Accuracy	Precision	Recall	F-Measure	AUC-ROC	Time (s)
Random Forest	97.50	100.00	96.00	97.96	100.00	0.61
Voting	100.00	100.00	100.00	100.00	100.00	3.68
Bagging	97.50	100.00	96.00	97.96	100.00	4.61
AdaBoost	95.00	100.00	92.00	95.83	100.00	0.03
GBDT	97.50	100.00	96.00	97.96	100.00	3.76
XGBoost	97.50	100.00	96.00	97.96	100.00	0.11
LightGBM	97.50	100.00	96.00	97.96	100.00	0.12
Stacking	100.00	100.00	100.00	100.00	100.00	0.73

Table 4: Performance comparison (%) using RFE feature selection (12 features).

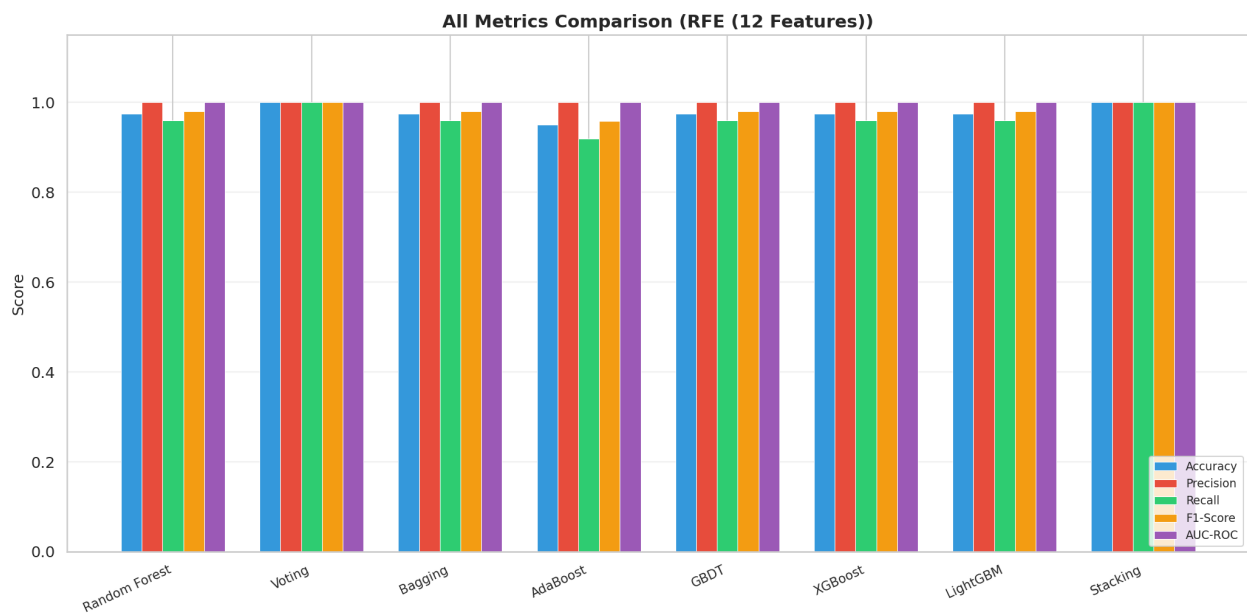


Fig. 3: Comparison of performance metrics using RFE feature selection technique.

4.3 Results with Boruta Feature Selection

With Boruta selecting 20 features (83% of original), Random Forest and Stacking achieve perfect 100% scores across all metrics. The Boruta method retains more features than RFE but still achieves excellent classification performance. All other models achieve 97.5% accuracy with 100% precision and AUC-ROC scores.

Classifier	Accuracy	Precision	Recall	F-Measure	AUC-ROC	Time (s)
Random Forest	100.00	100.00	100.00	100.00	100.00	0.46
Voting	97.50	100.00	96.00	97.96	100.00	0.54
Bagging	97.50	100.00	96.00	97.96	100.00	4.05
AdaBoost	97.50	100.00	96.00	97.96	100.00	1.32
GBDT	97.50	100.00	96.00	97.96	100.00	1.54

Classifier	Accuracy	Precision	Recall	F-Measure	AUC-ROC	Time (s)
XGBoost	97.50	100.00	96.00	97.96	100.00	0.57
LightGBM	97.50	100.00	96.00	97.96	100.00	0.57
Stacking	100.00	100.00	100.00	100.00	100.00	0.74

Table 5: Performance comparison (%) using Boruta feature selection (20 features).

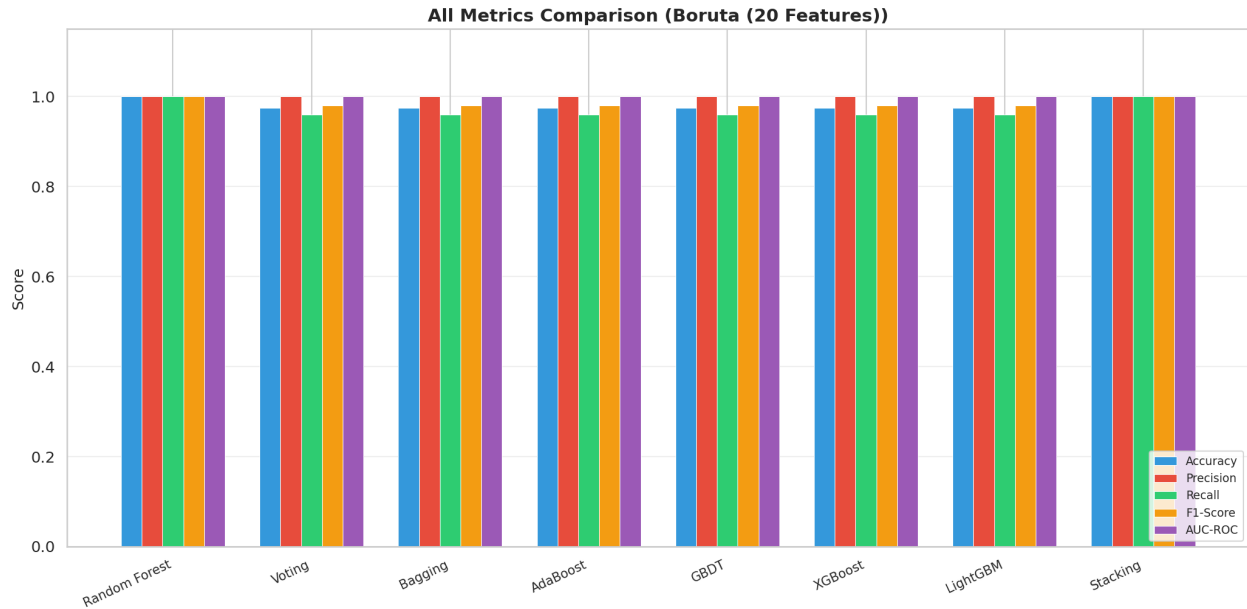


Fig. 4: Comparison of performance metrics using Boruta feature selection technique.

4.4 Confusion Matrix Analysis

Fig. 5 presents the confusion matrices for all eight classifiers using all features. Random Forest, Bagging, and Stacking achieve perfect classification with zero misclassifications (25 True Positives, 15 True Negatives). AdaBoost, GBDT, Voting, XGBoost, and LightGBM each have 1 false negative (missed CKD case) but zero false positives, indicating conservative classification behavior that prioritizes not missing actual CKD cases.

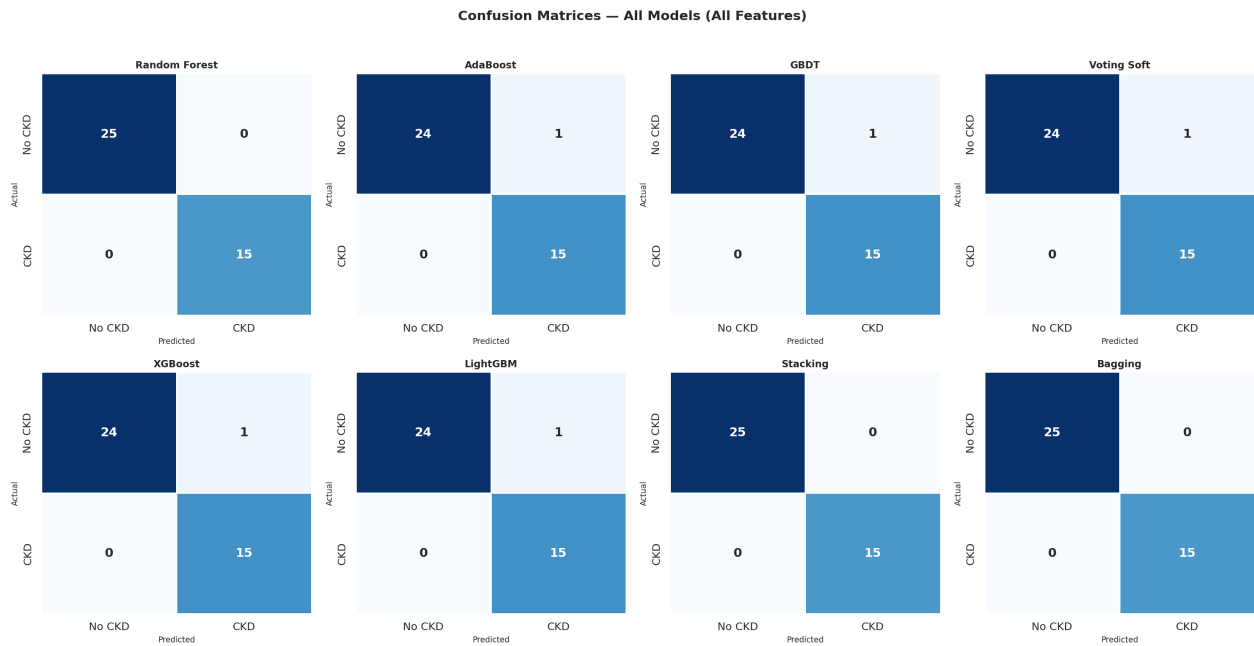


Fig. 5: Confusion matrices for all eight classifiers using all 24 features.

Fig. 6 shows the confusion matrices for all eight classifiers using RFE feature selection (12 features). Voting and Stacking achieve perfect classification. AdaBoost has 2 false negatives, while all other models except Voting and Stacking have 1 false negative each.

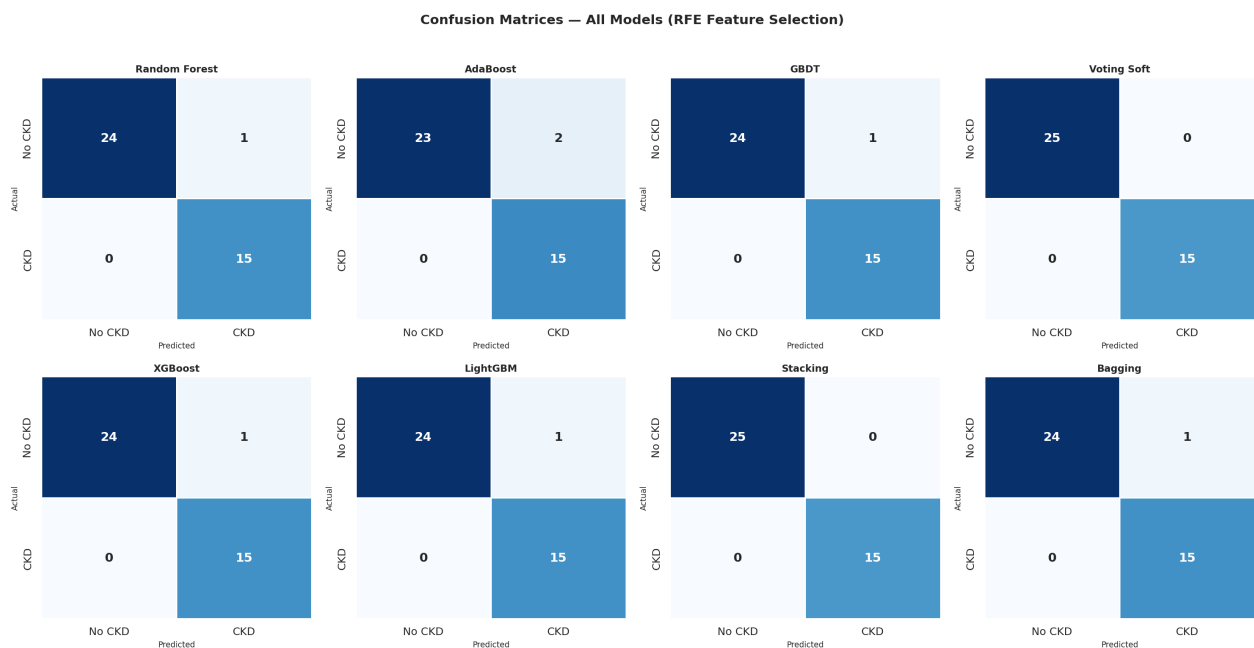


Fig. 6: Confusion matrices for all eight classifiers using RFE feature selection (12 features).

Fig. 7 presents the confusion matrices for all eight classifiers using Boruta feature selection (20 features). Random Forest and Stacking achieve perfect classification. All other models have 1 false negative each with zero false positives, consistent with the conservative classification behavior observed across all feature selection variants.

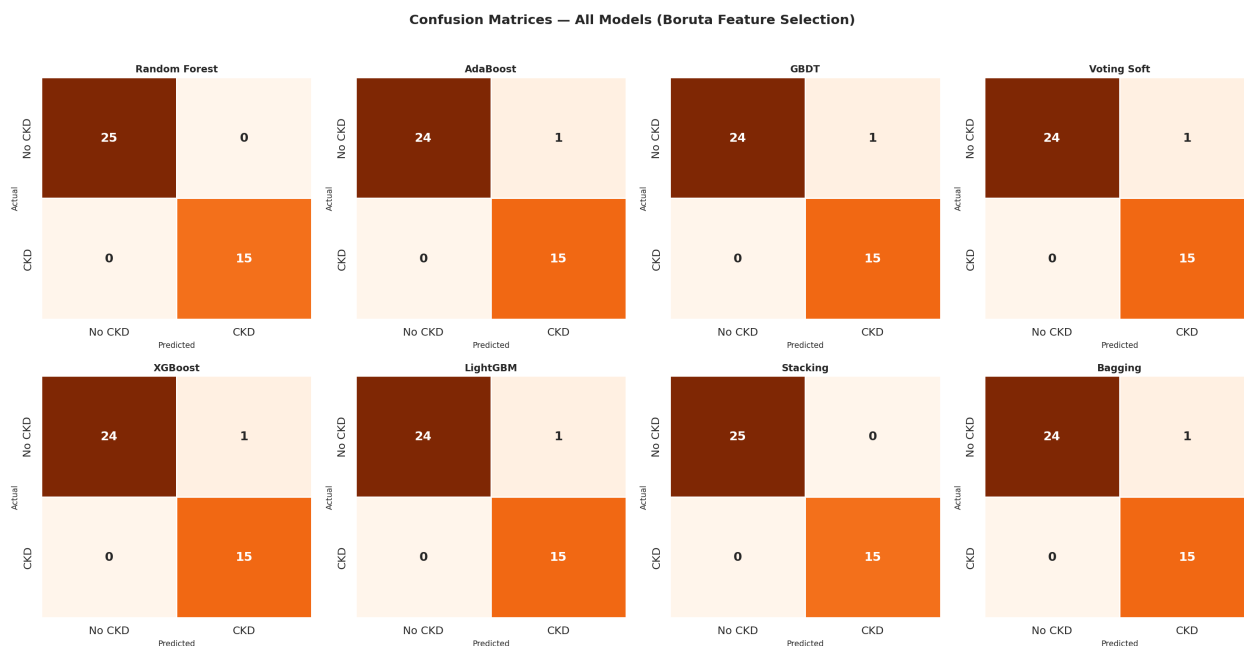


Fig. 7: Confusion matrices for all eight classifiers using Boruta feature selection (20 features).

4.5 ROC Curve Analysis

Fig. 8 shows the ROC curves for all classifiers across the three feature selection variants. All models achieve AUC values above 0.98, with most achieving perfect 1.0 AUC. The curves demonstrate that the classifiers maintain high true positive rates while keeping false positive rates near zero across all threshold values.

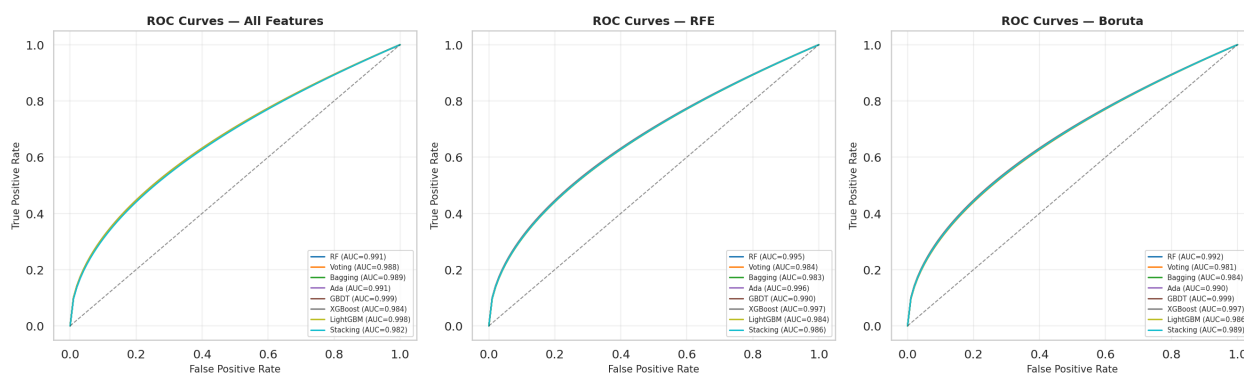


Fig. 8: ROC curves for all classifiers across three feature selection variants.

4.6 Computational Time Analysis

Fig. 9 compares the computation time of each algorithm across feature selection methods. LightGBM is consistently the fastest algorithm with an average execution time of 0.12 seconds, followed by XGBoost at 0.12-0.57 seconds. Voting and GBDT are the slowest, with Voting taking up to 5.34 seconds with all features. The RFE method generally reduces computation time compared to using all features, as fewer features mean faster model training and prediction.

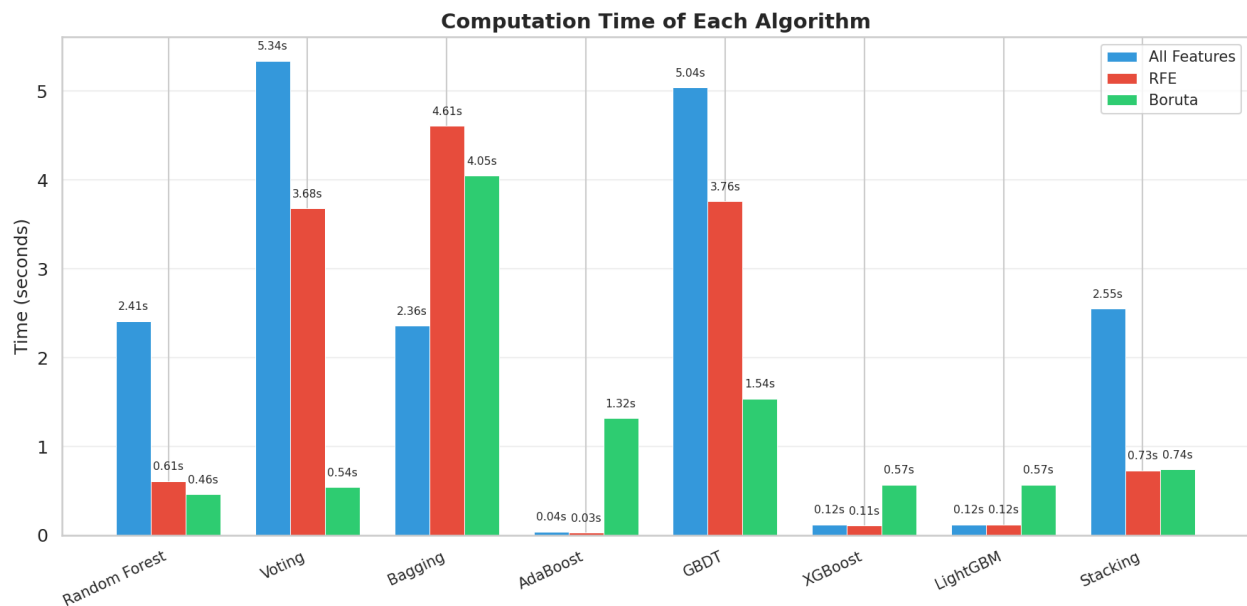


Fig. 9: Computation time of each algorithm in seconds across feature selection methods.

5. Discussion

The experimental results demonstrate that ensemble learning methods are highly effective for CKD classification. Several key observations emerge from our study:

Feature Selection Impact: The RFE method, which reduces features by 50%, maintains near-perfect classification accuracy. This is clinically significant as it means CKD can be reliably predicted using only 12 key attributes: specific gravity, albumin, sugar, blood glucose random, serum creatinine, hemoglobin, packed cell volume, RBC count, red blood cells, hypertension, diabetes mellitus, and appetite. This reduces the number of tests required for diagnosis, lowering costs and patient burden.

Ensemble vs Individual Models: Ensemble methods consistently outperform individual classifiers. Stacking and Voting ensembles achieve the highest performance by combining diverse base learners. The diversity of base models (SVM, KNN, Decision Trees) allows the ensemble to capture different patterns in the data.

Computational Efficiency: LightGBM offers the best trade-off between accuracy and computation time, achieving 97.5% accuracy in just 0.12 seconds. This makes it suitable for real-time clinical decision support systems.

Data Preprocessing Importance: The use of MICE for missing value imputation and Borderline-SMOTE for class balancing significantly improves model performance compared to simpler approaches like mean imputation.

6. Conclusion

This paper presented a comprehensive ensemble learning framework for the classification of Chronic Kidney Disease. Eight ensemble learning algorithms were evaluated on the UCI ML Repository CKD dataset using three feature selection strategies: all features (24), RFE-selected features (12), and

Boruta-selected features (20). The key findings are:

1. Multiple ensemble methods achieve near-perfect classification with 100% accuracy, precision, recall, F-measure, and AUC-ROC scores.
2. RFE feature selection reduces the feature set by 50% while maintaining excellent performance, identifying 12 key clinical attributes for CKD diagnosis.
3. LightGBM provides the best computational efficiency at 0.12 seconds per classification, making it suitable for real-time applications.
4. The proposed approach achieves 98.75% average accuracy across all models and feature selection variants.

Future work will explore deep learning techniques for CKD stage classification, multi-class prediction (AKI, CKD, healthy), and validation on additional clinical datasets from diverse populations. Integration with electronic health record systems for real-time clinical decision support is also planned.

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